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Edge AI Plant Care Monitor

On-device species ID, health assessment & daily care advice

Arduino UNO Q · ECE 180 · Final Project Proposal

Team #4

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What We Are Proposing

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A self-contained device that photographs a plant, identifies its species, assesses its health, and fuses that with live soil sensors to give a **daily care recommendation** — all running on the Arduino UNO Q, no cloud.

Where the AI/ML runs (on the AUQ):

- Two TFLite CNNs execute **on the UNO Q Ubuntu MPU**: *species ID* and *health classification* (healthy / stressed / diseased).
- A fusion + rule layer combines model output with soil moisture & pH against per-species ideal ranges to generate the recommendation — also on-device.

Must haves

- On-device species classification
- On-device health classification
- Live soil moisture + pH read on STM32
- Fused daily care recommendation

Nice to haves

- Fine-tuning on our own plant photos
- Trend tracking / history dashboard
- Live display output (MIPI-DSI)
- Small on-device LLM for phrasing advice

Project Plan — Gantt Chart (Week 2 → Week 5)

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